

**Kathmandu University Department of Artificial Intelligence Course**  
**Title: Natural Language Processing (NLP)**  
**Course Code: AIAC 536**  
**Credit Hours: 3 (45 Hours)**  
**Course Type: Elective**  
**Program: MTech in AI.**  
**Total Credit Hours: 45 Hours**

## **Course Description**

This course provides a comprehensive introduction to Natural Language Processing (NLP), covering both foundational linguistic concepts and modern deep learning approaches. Students will learn classical NLP techniques, statistical methods, neural networks, transformer architectures, and Large Language Models (LLMs). The course emphasizes practical implementation using contemporary frameworks such as PyTorch, Hugging Face Transformers, and modern generative AI tools. Special attention is given to multilingual NLP, retrieval augmented generation (RAG), prompt engineering, responsible AI, and emerging trends in foundation models.

## **Course Objectives**

Upon successful completion of this course, students will be able to:

1. Understand the linguistic foundations of NLP, including morphology, syntax, and semantics.
2. Apply classical, statistical, and neural approaches to solve NLP problems.
3. Implement NLP pipelines using modern Python-based libraries and frameworks.
4. Understand transformer architectures and foundation language models.
5. Fine-tune and evaluate pretrained language models for downstream tasks.
6. Design applications using LLMs, prompt engineering, and Retrieval-Augmented Generation (RAG).
7. Analyze multilingual and low-resource language challenges, including Nepali NLP.
8. Evaluate NLP systems using standard metrics and interpretability techniques.
9. Assess ethical, societal, and environmental implications of modern NLP systems.

## **Unit 1: Foundations of NLP and Linguistics (5 Hours)**

- Introduction to NLP: history, evolution, and applications
- NLP pipeline and text preprocessing
- Morphology: tokenization, stemming, lemmatization
- Syntax: POS tagging, constituency and dependency parsing
- Semantics and pragmatics
- Named Entity Recognition (NER) and Word Sense Disambiguation (WSD)
- Rule-based NLP systems and limitations

## **Unit 2: Statistical NLP and Text Representation (5 Hours)**

- Statistical NLP fundamentals
- N-grams and language modeling
- TF-IDF and vector space models
- Word embeddings: Word2Vec, GloVe, FastText
- Similarity measures: cosine similarity and semantic similarity
- Distributional semantics and limitations

## **Unit 3: Neural Networks for NLP (5 Hours)**

- Introduction to deep learning for NLP
- Feedforward neural networks
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM) and GRU networks
- Sequence modeling and sequence-to-sequence learning
- Encoder-decoder architectures
- Backpropagation Through Time (BPTT)

## **Unit 4: Attention Mechanisms and Transformers (6 Hours)**

- Limitations of recurrent architectures
- Attention mechanisms
- Self-attention and scaled dot-product attention
- Transformer architecture
- Multi-head attention
- Positional encoding
- Encoder-only, decoder-only, and encoder-decoder transformers
- Computational advantages of transformers

## **Unit 5: Foundation Models and Large Language Models (6 Hours)**

- Pretraining and fine-tuning paradigms
- Masked language modeling and autoregressive modeling
- BERT, RoBERTa, GPT, T5, and LLaMA architectures
- Instruction tuning and alignment
- Transfer learning in NLP
- Emerging trends in foundation models

## **Unit 6: Generative AI, Prompt Engineering, and RAG (6 Hours)**

- Fundamentals of Generative AI
- Prompt engineering techniques

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought prompting
- In-context learning
- Retrieval-Augmented Generation (RAG)
- Vector databases and semantic search
- Hallucination and grounding in LLMs
- LLM applications in knowledge-intensive tasks

## **Unit 7: NLP Applications and Multilingual NLP (5 Hours)**

- Text classification and sentiment analysis
- Question answering systems
- Machine translation
- Text summarization (extractive and abstractive)
- Dialogue systems and conversational AI
- Multilingual transformers (mBERT, XLM-R)
- Cross-lingual transfer learning
- Nepali and Indic language NLP case studies

## **Unit 8: Evaluation, Explainability, and Responsible AI (4 Hours)**

- Evaluation metrics:
  - Accuracy
  - Precision, Recall, F1-score
  - BLEU
  - ROUGE
  - Perplexity
- Explainable NLP
- Attention visualization
- Feature attribution methods
- Model robustness and debugging
- Fairness and transparency

## **Unit 9: Ethics, Safety, and Emerging Trends (3 Hours)**

- Ethical issues in NLP and Generative AI
- Bias and fairness in language models
- Privacy and data governance
- AI safety and hallucination risks
- Environmental impact of large-scale model training
- Future directions:
  - Multimodal LLMs
  - Agentic AI

- AI assistants
- Student project presentations

## **Practical Laboratory Components (Suggested)**

### **Lab 1: Text Processing and Linguistic Analysis •**

- Tokenization, normalization, stemming, lemmatization •
- POS tagging and NER using spaCy and NLTK

### **Lab 2: Statistical NLP**

- TF-IDF implementation
- N-gram language modeling
- Similarity analysis using embeddings

### **Lab 3: Word Embeddings**

- Word2Vec and FastText
- Visualization of embeddings using t-SNE

### **Lab 4: Neural NLP**

- Sentiment classification using LSTM/GRU
- Sequence-to-sequence modeling

### **Lab 5: Transformer Models**

- Fine-tuning BERT for classification or NER
- Text generation using GPT-based models

### **Lab 6: Prompt Engineering**

- Zero-shot and few-shot prompting
- Chain-of-thought reasoning experiments

### **Lab 7: Retrieval-Augmented Generation**

- (RAG)** • Building a simple RAG pipeline
- Vector databases and semantic retrieval

### **Lab 8: Multilingual NLP**

- Nepali or Indic language POS tagging, NER, or classification •

Cross-lingual transfer learning using mBERT/XLM-R **Lab 9:**

### **Evaluation and Explainability**

- Computing BLEU, ROUGE, F1, and Perplexity
- Attention visualization and error analysis

## Mini Project (Suggested)

Students will develop an end-to-end NLP application such as:

- Chatbot or Question Answering System
- Sentiment Analysis Platform
- Retrieval-Augmented Knowledge Assistant •
- Multilingual NLP Application
- Text Summarization System
- Document Classification System

## Hour Distribution Summary

|              |                                             |                 |
|--------------|---------------------------------------------|-----------------|
| <b>Unit</b>  | <b>Topic</b>                                | <b>Hours</b>    |
| 1            | Foundations of NLP and Linguistics          | 5               |
| 2            | Statistical NLP and Text Representation     | 5               |
| 3            | Neural Networks for NLP                     | 5               |
| 4            | Attention Mechanisms and Transformers       | 6               |
| 5            | Foundation Models and LLMs                  | 6               |
| 6            | Generative AI, Prompt Engineering & RAG     | 6               |
| 7            | NLP Applications & Multilingual NLP         | 5               |
| 8            | Evaluation, Explainability & Responsible AI | 4               |
| 9            | Ethics, Safety & Emerging Trends            | 3               |
| <b>Total</b> |                                             | <b>45 Hours</b> |